Use of Immune Checkpoint Inhibitor Pembrolizumab in the Treatment of High-Risk, Early-Stage Triple-Negative Breast Cancer: ASCO Guideline Rapid Recommendation Update

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ASCO Rapid Recommendations Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the ASCO Guideline Methodology Manual. The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options.

BACKGROUND

In 2021, ASCO published a guideline on neoadjuvant chemotherapy, endocrine therapy, and targeted therapy for breast cancer.[1] Recently, results of the fourth interim analysis of data from the KEYNOTE-522 randomized (2:1), double-blind, placebo-controlled trial were published.[2] KEYNOTE-522 evaluated the combination of neoadjuvant carboplatin and paclitaxel followed by doxorubicin or epirubicin and cyclophosphamide with pembrolizumab (n = 784) or placebo (n = 390), with completion of one year of pembrolizumab or placebo after surgery, in 1,174 patients with previously untreated, stage II or stage III triple-negative breast cancer (TNBC). Patients were eligible for KEYNOTE-522 regardless of programmed death-ligand 1 expression. Adjuvant capecitabine use was not allowed. The statistically significant event-free survival (EFS) results of KEYNOTE-522 trial constituted a strong signal for an update of the 2021 ASCO guideline recommendation on the role of pembrolizumab in the treatment of high-risk, early-stage TNBC.

METHODS

A targeted electronic literature search was conducted to identify phase III clinical trials pertaining to the recommendation on immune checkpoint inhibitors in this patient population. No additional randomized trials were identified. The original Expert Panel was reconvened to review the key evidence from KEYNOTE-522 and to review and approve the revision to the recommendation.

EVIDENCE REVIEW

With a median follow-up of 39.1 months, the updated analysis showed a statistically significant improvement in EFS among patients who received pembrolizumab plus neoadjuvant chemotherapy, followed by adjuvant pembrolizumab after surgery, compared with patients who received control therapy (3-year EFS 84.5% *v* 76.8%; hazard ratio 0.63, 95% CI, 0.48 to 0.82; $P <$ .001). At 36 months, the estimated overall survival (OS) in the pembrolizumab-chemotherapy group was 89.7% (95% CI, 87.3 to 91.7); the estimated OS in the placebo-chemotherapy group was 86.9% (95% CI, 83.0 to 89.9). Eighty patients (10.2%) in the pembrolizumab-chemotherapy group died, and 55 patients (14.1%) in the placebo-chemotherapy group died (hazard ratio for death, 0.72; 95% CI, 0.51 to 1.02). Of note, OS data—presented solely for descriptive purposes—were immature for this planned interim analysis and follow-up is ongoing.[2] Across the neoadjuvant and adjuvant phases of the trial, treatment-related adverse events (TRAEs) that were grade 3 or higher occurred in 77.1% and 73.3% of the patients in the pembrolizumab-chemotherapy group and in the placebo-chemotherapy group, respectively. Most adverse events occurred in the neoadjuvant treatment phase versus the adjuvant phase. The most commonly occurring grade 3 or higher TRAEs in the pembrolizumab-chemotherapy and placebo-chemotherapy groups were neutropenia (34.5% *v* 33.4%), neutrophil count decrease (18.6% *v* 23.1%), and anemia (18.0% *v* 14.9%). Four deaths in the pembrolizumab-chemotherapy group and one death in the placebo-chemotherapy group were attributed to TRAEs. TRAEs that led to discontinuation of the trial regimen occurred in 27.7% of patients in the pembrolizumab-chemotherapy group and 14.1% of patients in the placebo-chemotherapy group. Grade 3 or higher immune-mediated adverse events (irAEs) occurred in 12.9% of patients in the pembrolizumab-chemotherapy group and 1.0% of patients in the placebo-chemotherapy group. There was a higher

ASSOCIATED CONTENT

The companion to this article was published in the May 1, 2021 issue of *Journal of Clinical Oncology*.

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Accepted on March 4, 2022 and published at ascopubs.org/journal/jco on April 13, 2022: DOI https://doi.org/10.1200/JCO.22.00503

Evidence-Based Medicine Committee approval: February 28, 2022

incidence of any-grade endocrine disorders—hypo- or hyperthyroidism, adrenal insufficiency, thyroiditis, and hypophysitis—seen in the pembrolizumab-chemotherapy group (26.8%) than in the placebo-chemotherapy group (9.1%).

2022 UPDATED RECOMMENDATION

For patients with T1cN1-2 or T2-4N0 (stage II or III), early-stage TNBC, the Panel recommends use of pembrolizumab (200 mg once every 3 weeks or 400 mg once every 6 weeks) in combination with neoadjuvant chemotherapy, followed by adjuvant pembrolizumab after surgery. Adjuvant pembrolizumab may be given either concurrent with or after completion of radiation therapy. Given that irAEs associated with pembrolizumab therapy can be severe and permanent, careful screening for and management of common toxicities are required. The ASCO guideline for management of irAEs in patients treated with immune checkpoint inhibitor therapy offers detailed practice recommendations and should be consulted by clinicians who prescribe pembrolizumab for patients with early-stage TNBC, https://ascopubs.org/doi/full/10.1200/JCO.21.01440 (Type: Evidence based, benefits outweigh harms; Evidence quality: Intermediate; Strength of recommendation: Moderate).

QUALIFYING STATEMENTS

Results from KEYNOTE-522 are based on continued pembrolizumab in the adjuvant setting. There is uncertainty concerning the optimal adjuvant treatment given independent benefits of capecitabine in TNBC and olaparib in patients with germline *BRCA* mutations without pembrolizumab. There are no data to support the use of pembrolizumab in combination with either capecitabine or olaparib.

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GUIDELINE AND CONFLICTS OF INTEREST

The Expert Panel was assembled in accordance with ASCO's Conflict of Interest Policy Implementation for Clinical Practice Guidelines ("Policy," found at http://www.asco.org/guideline-methodology). All members of the Expert Panel completed ASCO's disclosure form, which requires disclosure of financial and other interests, including relationships with commercial entities that are reasonably likely to experience direct regulatory or commercial impact as a result of promulgation of the guideline. Categories for disclosure include employment; leadership; stock or other ownership; honoraria, consulting or advisory role; speaker's bureau; research funding; patents, royalties, other intellectual property; expert testimony; travel, accommodations, expenses; and other relationships. In accordance with the Policy, the majority of the members of the Expert Panel did not disclose any relationships constituting a conflict under the Policy.

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EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/breast-cancer-guidelines.

EQUAL CONTRIBUTION

D.L.H. and L.A.K. were Expert Panel cochairs.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI https://doi.org/10.1200/JCO.22.00503.

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ACKNOWLEDGMENT

The Expert Panel would like to thank Drs Aki Morikawa and Praveen Vikas and the ASCO Evidence Based Medicine Committee for their thoughtful reviews and insightful comments on this guideline. The following are members of the ASCO Neoadjuvant Chemotherapy, Endocrine Therapy, and Targeted Therapy for Breast Cancer Guideline Expert Panel: Lisa A. Carey, MD; Jennie R. Crews, MD; Dawn L. Hershman, MD; E. Shelley Hwang, MD; Seema A. Khan, MD; Larissa A. Korde, MD; Sibylle Loibl, MD, PhD; Elizabeth A. Morris, MD; Alejandra Perez, MD; Meredith M. Regan, ScD; Patricia A. Spears, BS; Preeti K. Sudheendra, MD; W. Fraser Symmans, MD; and Rachel L. Yung, MD. Finally, the Panel is grateful to Brittany E. Harvey for her assistance in completing the evidence summary for this rapid update.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Consulting or Advisory Role: AIM Specialty Health

No other potential conflicts of interest were reported.